Software Implementation Planning — roadmap

TOTAL: orig ~59d · forecast ~54d | PROGRESS: spent ~15.5d · rem ~38.5d | ~1d/week · +25% reserve

1

DONE
12 Sep 2026

Architecture baseline

TOTAL: orig ~1d · total ~0.7d

RESULT

- First software architecture baseline.
- TimingNode and application ownership are clear.
- Implementation can start deliberately.

DEMO

- Walk through the architecture diagrams.
- Explain TimingNode and application ownership.
- Show where the first executable fits.

DOCUMENT STATUS

10-SDP R-55 | 11-SIP R-60 | 30-UC O-25 | 31-SSSD W-45 | 60-SVP W-30

2

DONE
13 Sep 2026

Application core repository skeleton

TOTAL: orig ~6d · total ~1.8d

RESULT

- Application-core and runnable application artifacts.
- Clean Linux and Windows build/test.
- Traceable build identity and lifecycle.

DEMO

- Build from a clean checkout.
- Produce both artifacts.
- Start, identify and stop the application.

DOCUMENT STATUS

30-UC W-40 | 50-SDE-02 A-100 | 41-01-SSD R-25 | 43-01-SDD-02 W-55 | 32-03-IDD R-45 | 60-SVP W-35

3

DONE
29 Sep 2026

Application and API foundation

TOTAL: orig ~3d · total ~8.2d

RESULT

- SI-01 runs from external configuration.
- Public interfaces share application semantics.
- Black-box testing is repeatable.

DEMO

- Start SI-01 from configuration.
- Inspect version/status through public interfaces.
- Inspect IF-03 with the JavaFX test client.

DOCUMENT STATUS

30-UC W-55 | 41-01-SSD R-35 | 32-03-IDD R-65 | 32-11-IDD R-55 | 60-SVP W-45

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4

DONE
04 Oct 2026

First registration-system slice

TOTAL: orig ~6d · total ~5.1d

RESULT

- One TimingNode owns a durable, ordered registration stream.
- The same registration behaviour is available through the public API and Development Client.
- Restart rebuilds local registration history without inventing new records.

DEMO

- Open the TimingNode with a LocationId and submit one accepted registration.
- Observe the committed record in history and as a live update.
- Close, restart and confirm that the committed history is recovered.

DOCUMENT STATUS

03-DOM W-80 · 30-UC R-70 · 41-01-SSD R-55 · 32-03-IDD R-70 · 60-SVP W-60

5

DONE
09 Oct 2026

Simulated antenna input and runtime behaviour

TOTAL: orig ~6d · total ~0d

RESULT

- Simulated antenna observations reach the normal registration path.
- Normal manual-registration ADD is available through IF-03 with client-supplied
- Restart/recovery works with the same simulated input path used by automated tests.

DEMO

- Open one TimingNode and show the committed OPEN TimingData record.
- Run a short simulated-tag batch and show one representative two-tag passage reaching
- Inspect the runtime counters, restart SI-01 and continue using the same simulated input

DOCUMENT STATUS

30-UC R-80 · 41-01-SSD R-70 · 32-03-IDD R-80 · 60-SVP R-70

6

DONE
13 Oct 2026

Engineering Desktop Client

TOTAL: orig ~4d · total ~0d

RESULT

- SI-02 is a reusable Engineering Desktop Client software item.
- Java 21 / BentoFX / Windows app-image operation is CI-qualified.
- SI-02 design and user guidance are separated into SSD/SDD/SUM roles.

DEMO

- Start packaged SI-02 and connect it to a running SI-01.
- Exercise registration/simulation and one Lua engineering script.
- Show Device Log, Client Log and Remote Shell as independent engineering surfaces.

DOCUMENT STATUS

30-UC R-85 · 41-02-SSD R-70 · 43-02-SDD-01 W-80 · 44-02-GPD W-90 · 50-SDE-03 R-90 · 70-02-SUM W-70 · 32-03-IDD R-85 · 60-SVP R-80

Software Implementation Planning — roadmap

TOTAL: orig ~59d · forecast ~54d | PROGRESS: spent ~15.5d · rem ~38.5d | ~1d/week · +25% reserve

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ACTIVE

end 21 Dec 2026

Backoffice and multi-node integration

TOTAL: orig ~7d · forecast ~8.5d
PROGRESS: spent ~0d · rem ~8.5d

RESULT

- Multiple TimingNodes retain independent state, sequence and storage.
- Local registration continues during upstream outage.
- Engineering-API and RabbitMQ transports exercise the same semantic upstream contract.

DEMO

- Start a multi-node SI-01 test composition.
- Use the SI-02 Backend pane over IF-03 to send RaceData/StageStartTimes and inspect typed outbound...
- Switch the SI-02 Backend Simulator to RabbitMQ transport and repeat the recovery flow through the hosted...

DOCUMENT STATUS

30-UC R-90 | 41-01-SSD R-85 | 32-06-ISD W-55 | 33-06-IDD W-45 | 43-01-SDD-03 W-70 | 60-SVP R-85

8

PLANNED

end 25 Jan 2027

Target platform decision

TOTAL: orig ~4d · forecast ~4d

RESULT

- One target-platform direction is selected with its main risks understood.
- Required prototype hardware and any custom-board need are explicit.
- Step 9 can start without reopening the basic platform choice.

DEMO

- Compare the credible platform options against the known software/device needs.
- Show the selected hardware block diagram.
- Show the prototype parts/cost path and remaining platform risks.

DOCUMENT STATUS

10-SDP R-75 | 11-SIP R-80 | 50-SDE-01 W-60 | 60-SVP R-85

9

PLANNED

end 05 Apr 2027

Target bring-up and deployment proof

TOTAL: orig ~4d · forecast ~8d

RESULT

- A blank/clean selected target can be provisioned repeatably from documented inputs.
- Target resource limits are based on measurements rather than desktop assumptions.
- The image/provisioning, deployment/start/restart and recovery paths are known before

DEMO

- Start from the documented target image/provisioning baseline and bring up a clean target.
- Connect the Engineering Client through IF-03 and run a representative
- Show target measurements, persistence survival and a clean service/system restart.

DOCUMENT STATUS

50-SDE-01 R-75 | 60-SVP R-88 | 70-01-SUM W-55

Software Implementation Planning — roadmap

TOTAL: orig ~59d · forecast ~54d | PROGRESS: spent ~15.5d · rem ~38.5d | ~1d/week · +25% reserve

10

PLANNED

end 28 Jun 2027

Real timing-device integration

TOTAL: orig ~10d · forecast ~10d

RESULT

- Real devices feed the same application/domain paths already proven with simulation.
- Device health and recovery are observable.
- Simulated tests remain usable as the fast regression baseline.

DEMO

- Run a representative real-device registration/input flow.
- Observe the resulting state/data through supported public presentation paths,
- Disconnect or reset one device and demonstrate the supported recovery behaviour.

DOCUMENT STATUS

30-UC R-95 | 41-01-SSD R-95 | 60-SVP R-95 | 70-01-SUM W-70

11

PLANNED

end 06 Sep 2027

Integrated system and field proof

TOTAL: orig ~8d · forecast ~8d

RESULT

- The representative integrated setup can run and recover from selected failures.
- Operators can observe the important system state through normal interfaces.
- Remaining hardening work is based on field/integration evidence.

DEMO

- Run a representative timing session from input through GUI/backoffice.
- Interrupt one external dependency or device.
- Recover and finish the session without losing committed timing history.

DOCUMENT STATUS

30-UC A-95 | 41-01-SSD A-95 | 60-SVP A-95 | 70-01-SUM A-90 | 11-SIP R-95